

SHANNON, Australia

WMO: 956170

Lat: 34.5683S

Lon: 116.3367E

Elev: 210

StdP: 98.83

Time Zone: 8.00 (E08)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.6	4.6	2.4	4.6	6.9	3.3	4.9	7.9	8.6	12.4	8.1	13.0	1.3	0	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.5	31.6	19.2	29.3	18.7	27.2	18.2	20.7	27.8	19.8	26.3	19.0	24.9	4.3	0

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.0	21.5	18.0	13.3	21.0	17.2	12.6	20.5	60.7	27.8	57.5	26.3	55.0	25.0	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	7.0	6.4	DB	1.3	37.0	1.0	1.5	0.6	38.1	0.0	38.9	-0.6	39.7	-1.3	40.8	(4)
			WB	1.2	22.9	1.2	1.3	0.3	23.8	-0.4	24.6	-1.0	25.3	-1.9	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	18.7	19.2	18.0	15.7	13.3	11.4	10.5	10.8	11.3	13.0	15.6	17.1
	DBStd	4.12	3.04	3.27	3.42	2.94	2.35	1.87	1.84	2.03	2.22	2.65	3.29	3.30
	HDD10.0	52	0	0	0	1	1	7	16	14	11	3	1	0
	HDD18.3	1573	33	26	48	91	156	207	243	234	211	166	97	63
	CDD10.0	1708	268	259	249	171	105	50	31	38	51	97	168	221
	CDD18.3	186	43	51	38	11	1	0	0	0	2	14	26	
	CDH23.3	1661	397	428	308	65	5	0	0	0	2	22	156	278
	CDH26.7	569	140	152	111	11	0	0	0	0	2	51	103	
Wind	WSAvg	3.3	3.7	3.6	3.5	2.9	2.7	3.0	3.1	3.2	3.3	3.3	3.4	3.5
Precipitation	PrecAvg	1062	18	20	37	68	124	161	187	150	116	74	50	28
	PrecMax	1271	75	57	107	149	208	260	273	223	180	120	138	70
	PrecMin	826	4	0	4	8	70	76	79	52	39	38	10	5
	PrecStd	132	18	15	25	40	40	54	50	39	40	27	31	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	33.5	33.9	28.2	23.8	18.7	17.9	19.7	22.4	26.1	31.5	33.8
		MCWB	19.9	20.0	19.5	17.6	15.6	14.2	13.9	13.4	14.4	17.2	17.8	18.4
	2%	DB	30.2	30.7	29.8	25.2	20.9	17.3	16.2	17.5	19.2	22.6	27.7	29.4
		MCWB	19.2	19.5	19.1	17.1	15.2	13.6	12.9	13.4	14.0	16.1	17.7	18.6
	5%	DB	27.7	28.3	26.6	22.8	19.1	16.2	15.1	15.9	17.3	20.1	24.5	26.2
		MCWB	18.4	18.7	18.4	16.3	14.9	13.3	12.5	12.7	13.2	15.1	16.8	17.7
	10%	DB	25.1	25.8	24.0	20.7	17.6	15.1	14.1	14.6	15.7	18.0	21.7	23.5
		MCWB	17.7	18.4	17.4	15.8	14.4	12.8	11.9	12.0	12.6	14.0	16.0	16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	22.8	21.9	19.1	18.0	16.0	15.2	15.5	16.2	18.6	19.9	20.8
		MCDB	28.9	28.4	28.4	23.9	20.6	17.5	16.4	17.3	19.2	23.7	27.6	29.0
	2%	WB	20.2	20.7	20.2	18.1	16.6	14.8	14.0	14.1	14.7	17.1	18.6	19.3
		MCDB	27.6	27.6	26.1	22.2	19.1	16.1	15.3	16.3	17.4	21.4	24.5	26.7
	5%	WB	19.2	19.7	19.2	17.2	15.8	14.0	13.0	13.2	13.7	15.7	17.5	18.3
		MCDB	26.0	25.7	24.2	20.9	18.0	15.5	14.4	15.1	16.3	19.0	22.9	25.0
	10%	WB	18.3	18.8	18.3	16.5	15.0	13.1	12.2	12.5	12.8	14.5	16.5	17.2
		MCDB	23.9	24.4	22.3	19.9	17.1	14.7	13.7	14.2	15.1	17.5	21.4	23.0
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.5	10.5	9.4	8.3	7.2	7.0	7.4	8.2	9.2	10.6	11.3
		MCDBR	16.8	16.0	14.9	12.6	10.5	8.9	8.6	9.4	11.2	13.3	15.7	16.5
		MCWBR	6.6	5.8	5.5	5.5	5.6	5.9	5.6	5.5	6.2	7.1	6.8	6.5
	5% WB	MCDBR	14.4	13.9	12.2	10.3	8.3	7.0	6.7	7.8	9.5	12.0	13.4	14.4
		MCWBR	6.3	5.8	5.2	5.3	5.4	5.4	5.1	5.3	6.0	6.8	6.6	6.3
Clear-Sky Solar Irradiance	taub	0.328	0.327	0.317	0.310	0.305	0.300	0.298	0.305	0.329	0.322	0.327	0.322	
	taud	2.594	2.608	2.632	2.647	2.627	2.620	2.624	2.596	2.501	2.545	2.557	2.595	
	Ebn at Noon	1009	990	963	916	865	842	864	906	933	983	1003	1018	
	Edh at Noon	104	98	89	78	70	66	69	81	100	104	107	104	
All-Sky Solar Radiation	RadAvg	7.47	6.55	5.18	3.72	2.73	2.22	2.40	3.17	4.25	5.51	6.65	7.39	
	RadStd	0.49	0.33	0.36	0.22	0.20	0.14	0.10	0.16	0.20	0.36	0.41	0.42	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-79	N/A	N/A

Nomenclature: See separate page